COM435 Software Development Practice
Sprint Specification

Team Number: 50

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Groomed Sprint Backlog

Priority	Story ID	Task ID	Task Name
1	S1	T1	Add database table for workouts.
2	S1	T2	Build a basic HTML workout page.
3	S1	T3	CSS styling of workout page.
4	S1	T4	HTML code to display a workout video beside each workout
5	S1	T5	Write C# server side code using ASP.NET Core Razor Page framework to get the workouts stored in database and display them on the HTML page (view)
6	S2	T6	Design Data Model
7	S2	T7	Build a basic HTML meal planner page
8	S2	T8	Build allergy restrictions HTML page
9	S2	T9	Add custom CSS for HTML pages
10	S2	T10	Write C# Server Side code for processing meal search results
11	S6	T11	Add database table for storing user calories
12	S6	T12	Build HTML calorie budget page
13	S6	T13	Write C# Server Side code for processing calories
14	S6	T14	Add custom CSS for calorie budget page
15	S6	T15	Form validation for calorie inputs
16	S11	T16	Design data model to hold personal trainer info
17	S11	T17	Build HTML for “Create coaching platform” page
18	S11	T18	Custom CSS styling for “Create coaching platform” page
19	S11	T19	Form validation for Personal Trainer data input
20	S13	T20	Build stopwatch user interface with HTML and CSS

Development Tasks for Story ID: S1

Browse workout library to see a list of workouts to participate in.

Task ID: T1	**Task name:** Add database table for workouts.
Task size	Medium
Dependencies	
Checklist of subtasks	• Create a database using a relational DBMS e.g., MYSQL or Microsoft SQL server • Define a Primary Key to uniquely identify workouts e.g., Workout ID • Define the workout attributes (table columns) such as name of workout, category/type of workout, duration, intensity, difficulty level, equipment needed, sets/reps and workout instructions. • Ensure suitable data types for each column. • Store workout videos in a File system and include the relevant file path in the workout table for each workout record. • Ensure Third Normal Form.

Task ID: T2	**Task name:** Build a basic HTML workout page.
Task size	Medium
Dependencies	• T1
Checklist of subtasks	• Set up basic HTML page with header, navigation bar, main content and footer. • Group related workouts together under specific HTML headings <h2> e.g., Cardio, Strength, Stretching, Yoga, Rehabilitation etc. The HTML header tags will provide clear separation between the types of workouts available and allow the user to browse for what they are interested in. • Tooltips are provided to specify extra information to the user when they hover over each element in the list. • Incorporate accessibility options such as buttons for switching between dark/light mode, buttons for altering the size of the text on screen, and ensuring alt text for images is provided for users with screen readers.

Task ID: T3	**Task name:** CSS styling of workout page.
Task size	Medium
Dependencies	• T2
Checklist of subtasks	• Reference external CSS stylesheet in HTML <head> section of page • Ensure font size and colour are appropriate and text can be easily understood and read by users. • Use CSS media queries to apply different styling for small and large viewports to scale the workout list appropriately. • Ensure proper alignment of HTML elements on the page. • Minify CSS • Alternatively, developers could use Bootstrap to help with CSS styling. It includes predefined classes for easy layout options and a responsive, mobile first grid system that appropriately scales as the device or viewport size increases.

Task ID: T4	**Task name:** HTML code to display a workout video beside each workout
Task size	Medium
Dependencies	• T1 • T2 • T5
Checklist of subtasks	• Use <video> element to display videos on HTML page. • Get the relevant video file path from the database table to pass to the src attribute of the video element to display the appropriate video. • Ensure video controls are included such as play, pause, volume and the ability to expand the size of the video. • Videos should be prevented from playing automatically, only when the user requests them to play. • Subtitles should be included for each workout video for users with hearing difficulties. • Ensure video format is MP4 to ensure support across the major browsers like Chrome, Firefox and Safari.

Task ID: T5	**Task name:** Write C# server side code using ASP.NET Core Razor Page framework to get the workouts stored in database and display them on the HTML page (view)
Task size	Large
Dependencies	• T1 • T2
Checklist of subtasks	• IList Collection object declared that holds all the details of the workouts stored in the database • List all the details in the OnGetAsync() method using the .ToListAsync() method • Use @HTML.DisplayNameFor and @HTML.DisplayFor helper tags to display the results on the content page (view) • Iterate through the collection object using a foreach loop to get the stored values.

<u>Development Tasks for Story ID: S6</u>

Calorie Budget page to allow users to input their calorie intake throughout the day against a max recommended calorie count.

Task ID: T11	**Task name:** Add database table for storing user calories
Task size	Medium
Dependencies	
Checklist of subtasks	• Define a Primary Key to uniquely identify daily calorie intake e.g., Calorie Intake ID • Define a Foreign Key to link each calorie record in the calorie table with a user in the User table (User ID) • Define the calorie attributes (table columns) such as calorie budget, breakfast calories, lunch calories, dinner calories, total calorie consumption, and date. (Storing date will allow the user to see how well they have stuck to their calorie budget over a period of time-however this function will appear in a later release) • Ensure suitable data types for each column. • Ensure Third Normal Form.

Task ID: T12	**Task name:** Build HTML calorie budget page
Task size	Medium
Dependencies	• T11

Checklist of subtasks	• Build HTML page with header, navigation bar, main content and footer. • Build HTML form to allow users to set their calorie budget and enter their calorie intake at breakfast, lunch and dinner into text input fields. Submit button to submit the form data to controller for processing. • Check HTML standards using W3C Markup Validation Service

Task ID: T13	**Task name:** Write C# Server Side code for processing calories
Task size	Large
Dependencies	• T11 • T12 • T15
Checklist of subtasks	• Page Controller (PageModel file) should enable the processing of user input to calculate the combined calorie consumption across breakfast, lunch and dinner for the user. • Bind Property attribute set on page model properties to enable model binding. Properties should be of the correct data type (int) • Compare combined calorie consumption against calorie budget to evaluate whether the user has stayed under their budget or exceeded their daily calorie allowance. • Message sent back to user informing them if they have met their calorie budget and the amount of calories under or over the budget they were for that particular day. • Store inputted calories in database table.

Task ID: T14	**Task name:** Add custom CSS for calorie budget page
Task size	Medium
Dependencies	• T12
Checklist of subtasks	• Reference external CSS stylesheet in HTML <head> section of page • Use Bootstrap to ensure responsive web design so that elements automatically adjust for different screen sizes (desktops, tablets and phones) and viewports.

	• Check CSS standards using W3C CSS Validation Service.

Task ID: T15	**Task name:** Form validation for calorie inputs
Task size	Medium
Dependencies	• T12
Checklist of subtasks	• Ensure security against SQL injection attacks or other malicious data entry which could result in the compromise of data stored in the database. • Suitable error messages are designed which inform the user what they have done wrong when inputting their calories in an incorrect manner. • Set range validation on calorie inputs so the user cannot enter unsuitable calorie numbers such as -500 or 30,000. • Set required attribute on calorie inputs so the user cannot submit without providing a valid entry. • Appropriate validation so the user cannot enter letters or special characters into the form fields, only numbers.

Test Cases

Test Case ID: TC1			
Test Description: Navigate to Measured Runs page and start a timed run			
User Story ID: S13	**User Story Title:** Basic stopwatch to record the time of a run (in hours, mins, and seconds)		
Step ID	**Step Description**	**Test Data**	**Expected Result**
1	Browse to Measured Runs page		Measured Runs page is displayed and user can see button to start a timed run and timer initially set to 00:00:00.
2	User clicks on "Start" button to begin a timed run		Timer begins counting and is displayed to the user so they can see the time elapsed so far.

Test Case ID: TC2			
Test Description: Navigate to Measured Runs page and start and stop a timed run			
User Story ID: S13	**User Story Title:** Basic stopwatch to record the time of a run (in hours, mins, and seconds)		
Step ID	**Step Description**	**Test Data**	**Expected Result**
1	Browse to Measured Runs page		Measured Runs page is displayed and user can see button to start a timed run and timer initially set to 00:00:00.

2	User clicks on "Start" button to start a timed run		Timer begins incrementing and is displayed to the user so they can see the time elapsed so far.
3	User clicks on "Stop" button to stop timed run		Timer stops incrementing and the user can see the time they achieved their run in.

Test Case ID: TC3			
Test Description: Browse nutritional breakfast meals with no allergies or restrictions set			
User Story ID: S2	**User Story Title:** Browse nutritional meals that are organized into easy to navigate categories e.g., breakfast, lunch, dinner and dessert.		
Step ID	**Step Description**	**Test Data**	**Expected Result**
1	Browse to Meal Planner page		Meal Planner page is displayed to user with a search button and a "+" button beside "Allergies and restrictions" to allow user to screen the meals they see based on their selections.
2	Click search button and enter 'Breakfast' into search bar	Breakfast	Search entry is acceptable.
3	View matched results		Images of breakfast meals are displayed along with their name. All breakfast meals are displayed since no allergy restrictions have been applied by the user.

Test Case ID: TC4			
Test Description: Browse nutritional lunch meals with Peanut-Free allergy restriction set			
User Story ID: S2	**User Story Title:** Browse nutritional meals that are organized into easy to navigate categories e.g., breakfast, lunch, dinner and dessert.		
Step ID	**Step Description**	**Test Data**	**Expected Result**
1	Browse to Meal Planner page		Meal Planner page is displayed to user with a search button and a "+" button beside "Allergies and restrictions" to allow user to screen the meals they see based on their selections.
2	Click "+" button beside Allergies header to add allergy restrictions for meals		User redirected to page with list of possible allergy restrictions e.g., Dairy-Free, Gluten Free, Peanut-Free etc.
3	Click "Peanut-Free" from allergy list		Peanut-Free has a tick beside it in the list

4	Click on return arrow		User brought back to meal planner page. “Peanut-Free” is shown beneath the Allergy header as it has been selected in the previous step.
5	Click Search button and enter ‘Lunch’ into search bar	Lunch	Search entry is acceptable.
6	View matched results		Images of lunch meals are displayed along with their name. Only matching lunch meals that are peanut-free are displayed to the user.

Test Case ID: TC5			
Test Description: User can access workout library and can see a list of available workouts with appropriate details and instructions along with a workout video for demonstration.			
User Story ID: S1	**User Story Title:** Browse workout library to see a list of workouts to participate in.		
Step ID	**Step Description**	**Test Data**	**Expected Result**
1	Browse to workout library page		Workout Library page is displayed to user.
2	Scroll down the page to see a list of workouts.		Workouts are displayed to the user and are categorised under specific headings such as Cardio, Strength and Conditioning, Yoga, Rehabilitation etc.
3	View individual work out details		Each workout in the list includes the name of the workout, category/type of workout, duration, intensity, difficulty level, equipment needed, sets/reps and workout instructions along with a workout video performed by a Personal Trainer.

Test Case ID: TC6			
Test Description: Navigate to workout library page and play a workout video			
User Story ID: S1	**User Story Title:** Browse workout library to see a list of workouts to participate in.		
Step ID	**Step Description**	**Test Data**	**Expected Result**
1	Browse to workout library page		Workout Library page is displayed to user.
2	Click on a workout video.		Workout video starts playing with Personal Trainer giving a

			demonstration of how the chosen workout is performed.
3	Click on expand video button.		The video expands to the size of the user's screen.
4	Click on volume button and adjust volume level		The video volume adjusts depending on users setting.
5	Click on subtitles button		Subtitles appear at the bottom of the workout video.

Test Case ID: TC7			
Test Description: Enter a valid daily calorie intake number into the Calorie Budget form field			
User Story ID: S6	**User Story Title:** Calorie Budget page to allow users to input their calorie intake throughout the day against a max recommended calorie count.		
Step ID	**Step Description**	**Test Data**	**Expected Result**
1	Navigate to Calorie Budget page		Calorie Budget page displays and calorie budget form field has focus.
2	Manually enter valid calorie number	2400	Calorie Budget is accepted and submit button has focus.
3	Click submit button		Calorie Budget has been set at 2400 calories for each day.

Test Case ID: TC8			
Test Description: Message displayed to the user when they have exceeded their calorie budget			
User Story ID: S6	**User Story Title:** Calorie Budget page to allow users to input their calorie intake throughout the day against a max recommended calorie count.		
Step ID	**Step Description**	**Test Data**	**Expected Result**
1	Navigate to Calorie Budget page		Calorie Budget page displays and the calorie budget for the user is set at 2400 calories.
2	Enter calories consumed at breakfast into Breakfast form field	600	Breakfast calories is accepted and Lunch field has focus
3	Enter calories consumed at lunch into Lunch form field	1000	Lunch calories is accepted and Dinner field has focus
4	Enter calories consumed at dinner into Dinner form field	1000	Dinner calories is accepted.
5	Click "Submit calorie intake" button		A message is displayed to the user instructing them they have exceeded their daily calorie budget by 200 calories.

Test Case ID: TC9			
Test Description: Attempt to access tailored workout page from hub with active subscription to specific personal trainer			
User Story ID: S11	**User Story Title:** Personal Trainer can set up a tailored coaching page with personalised branding, e.g., logo, and information about their fitness expertise.		
Step ID	**Step Description**	**Test Data**	**Expected Result**
1	Navigate to personal trainer area from hub		Personal trainer area opens and app changes branding and colour scheme to match personal trainer branding.
2	Click Tailored Workouts		Tailored workouts page opens showing bespoke workouts chosen by PT specifically for that user.
3	Click specific exercise for more details	Click 'Barbell Bench Press'	Exercise details opens displaying exercise name, weight history of user, photo and video links showing example reps.

Test Case ID: TC10			
Test Description: Attempt to access tailored workout page from hub with an expired subscription to specific personal trainer			
User Story ID: S11	**User Story Title:** Personal Trainer can set up a tailored coaching page with personalised branding, e.g., logo, and information about their fitness expertise.		
Step ID	**Step Description**	**Test Data**	**Expected Result**
1	Navigate to personal trainer area from hub		Personal trainer area opens and app changes branding and colour scheme to match personal trainer branding
2	Click Tailored Workouts		The tailored workouts page does not open, instead the screen displays a padlock symbol and an error message saying, 'Your subscription has expired'. Links to resubscribe are highlighted.
3	Click specific exercise for more details	Click 'Barbell Bench Press'	No specific exercises are shown – content should be blocked until the user's subscription is active.